

Module 08: Validation-Based Protocol (OCC) and Phantom Phenomena

AKTU BCS501 | Unit 5: Concurrency Control Techniques

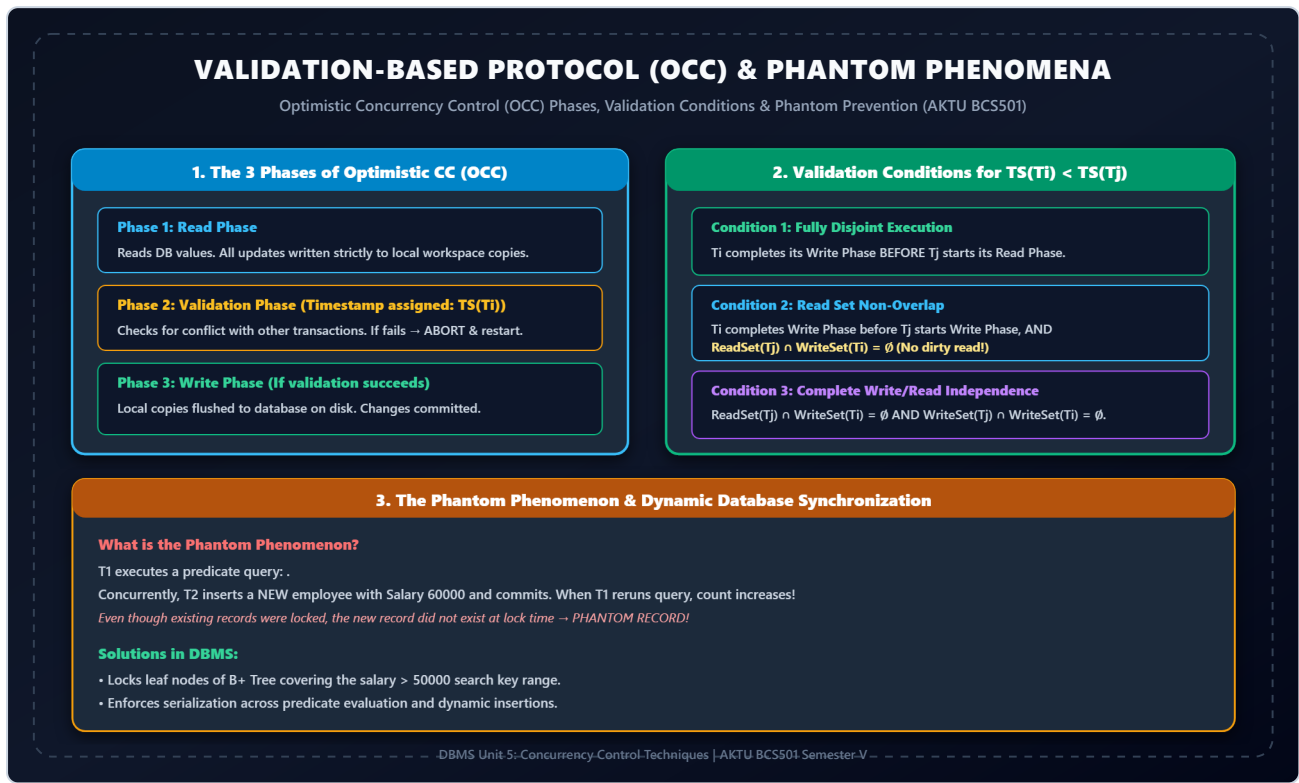


Figure 8.1: The 3 OCC Execution Phases, Validation Conditions & Phantom Prevention

1. The 3 Phases of OCC

- 1. Read Phase:** Reads DB and buffers updates locally in memory.
- 2. Validation Phase:** Checks conflict conditions; assigns TS(T).
- 3. Write Phase:** Commits updates to disk if validation passes.

2. The Phantom Phenomenon

Predicate queries (e.g., `SELECT COUNT(*) WHERE Salary > 50k`) can see phantom records if concurrent transactions insert new matching rows. Solved via **Index-Range Locking** on B+ Tree leaves or Predicate Locking.